

Do On-Chain Signals Improve Bitcoin Market-Regime Classification?

链上数据能否改善比特币市场阶段分类？

CPSC 381/581: Machine Learning | Yale University | Instructor: Alex Wong
(中英双语版 · Bilingual Reference Edition)

1. Name

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2. Problem Statement

Bitcoin markets cycle through three qualitative regimes recognized by practitioners and researchers: **Accumulation** (price consolidation, informed buyers absorbing supply), **Expansion** (sustained directional trend), and **Distribution** (price plateau, early holders reducing exposure). Correctly identifying the current regime has direct practical value for position sizing, risk management, and timing.

The central research question: *do on-chain features carry regime-predictive information beyond what price-derived features already encode?* This is framed as a supervised **multi-class classification** task: given a feature vector at time t , predict the regime label $y \in \{\text{Accumulation, Expansion, Distribution}\}$ over the next 30 days.

【问题背景】比特币市场有三个公认的周期阶段：

- 吸筹期 Accumulation：价格横盘，聪明资金在低位买入，链上老币不动
- 扩张期 Expansion：价格持续上涨，市场广泛参与
- 派发期 Distribution：价格见顶附近，早期持币者开始减仓卖出

核心研究问题：加上链上数据之后，对阶段的预测准确率会不会比只用价格数据更高？这是一个三分类监督学习问题：给定时间 t 的特征向量，预测接下来30天属于哪个阶段。

Why non-trivial / ■■■■■■■■■■

- Time series violate i.i.d. assumptions — serial correlation, distribution shift across cycles, severe class imbalance.
- The on-chain $\rightarrow$ regime mapping is indirect; a null result (no improvement) is equally valid scientifically.
- Labels must be constructed, not downloaded — this is a modeling choice with real consequences.

- 金融时间序列不满足独立同分布假设：数据有自相关性、跨周期分布会变、三个类别样本量严重不均衡（派发期只占约15%）
- 链上信号到市场阶段的映射关系是间接的——如果加了链上数据之后模型没有改善，这本身也是一个有价值的发现
- 标签（哪段时间是吸筹/扩张/派发）不是现成的，需要我们自己定义，这个选择本身就是建模决策的一部分

Acknowledged limitations / ■■■■■■■■■■

■ On-chain data only captures spot-market holder behavior. Short-term BTC volatility is largely driven by derivatives (perpetual futures, options) which leave no on-chain trace. This project targets medium-term regime identification (30-day), not short-term price prediction.

【重要局限】链上数据只能看到现货持币者的行为，但比特币短期价格波动很大程度上由衍生品市场（永续合约、期权）驱动，而这些在链上完全看不到。所以这个项目针对的是30天中期阶段识别，而不是短期价格预测。

■ If this model became widely known, rational traders would arbitrage away its signals. The contribution is the empirical test of signal existence, not a claim of persistent alpha.

【另一个局限】如果这个模型的信号被市场广泛使用，理性套利者会让信号失效。本研究的学术贡献是检验信号是否存在，而不是声称能持续赚钱。

3. General Approach

3.1 Label Construction — the hardest part

Ground-truth labels do not exist and must be constructed using two steps: (1) Rule-based initialization using MVRV ratio thresholds calibrated against historical cycle peaks/troughs. (2) Sensitivity analysis under three threshold variants — results robust across all three are considered credible.

■ **Critical:** MVRV is used both to construct labels AND as a feature. An ablation removing MVRV from features verifies other on-chain signals are independently predictive.

【标签构建——最难的部分】正确的市场阶段标签不是现成的，需要用两步来定义：

- 第一步：用 MVRV 比率设定阈值。 $MVRV < 1.0$ = 吸筹（持币者平均处于亏损状态）； $MVRV > 3.5$ = 派发（持币者平均浮盈巨大）；中间 = 扩张。阈值根据历史四个周期顶底校准。
- 第二步：用三组不同阈值重新生成标签，做敏感性分析。三组结论一致的结果才可信，结论随阈值翻转的结果要标注为【可能是人为产物】。

【重要设计】MVRV 既用来构建标签，又作为特征输入模型，存在循环论证风险。解决方案：做一次去掉 MVRV 的消融实验，验证其他链上指标单独是否有预测力。

3.2 Model Progression / ■■■■■■

Stage	Model	Course Topic	Purpose
1	Ridge Regression 30	Week 3/9 +	
2	Linear SVM	Week 11	Classification L2
3	Random Forest + XGBoost	Week 7/ensemble +	SHAP
4	1-D Temporal CNN PyTorch	Week 13 SGD	

3.3 Walk-Forward Expanding Window / ■■■■■■

Standard k-fold cross-validation leaks future data into training for time series. Instead: train on cycles 1–k, test on cycle k+1. With four Bitcoin halving cycles (2013–15, 2016–18, 2019–21, 2022–26), this gives three

non-overlapping test folds.

【为什么不用普通K折交叉验证】普通K折会把未来数据泄露到训练集里，对时间序列来说是作弊行为。正确方法是：用前N个周期训练，在第N+1个周期测试，滚动前进。四个减半周期 = 三个独立测试fold。

■ *Early folds have far fewer training samples (~300–500 days vs ~2,000). Per-fold variance is reported explicitly — not hidden in a single average.*

【诚实报告】早期fold训练样本少，模型稳定性差。每个fold单独报告，不做平均掩盖这个问题。

4. Data

Dataset: January 2015 – March 2026 (~4,100 daily observations), covering three complete Bitcoin halving cycles. Two freely accessible sources merged on UTC date index.

数据范围：2015年1月至2026年3月，约4,100个交易日，覆盖三个完整比特币减半周期。两个免费数据源按日期合并。

Source / 	Variables / 	Rationale / 	Access / 
CoinMetrics Community API	MVRV, Realized Cap, NVT, Coin Days Destroyed, Active Addresses, Hash Rate, Puell Multiple, SOPR		
Yahoo Finance (yfinance)	 OHLCV, 30  , RSI-14, MACD  , BTC 		yfinance Python 

Key ablation design: Model (A) = price only. Model (B) = price + on-chain. The F1 delta between B and A is the direct answer to the research question. A null or negative delta is reported honestly.

【关键实验设计】模型A只用价格特征，模型B用价格+链上特征。B比A高出的F1分数就是链上数据的边际贡献。如果B不比A好，这也是一个诚实且重要的结论，不算失败。

■ *SOPR and some on-chain metrics only have reliable coverage post-2018. Features with short history are excluded from early folds to avoid imputation bias.*

部分链上指标（如SOPR）2018年之前数据不可靠，早期fold中会把这些特征排除掉，避免插值填充引入偏差。

5. Related Work

① Bitcoin regime structure (Ardia, Bluteau & Rüede, 2019)

Applies Markov-Switching GARCH to Bitcoin daily returns, finding strong statistical evidence of at least two distinct volatility regimes. Single-regime GARCH models systematically underperform in VaR forecasting. Their work motivates this project: if Bitcoin is structurally regime-dependent, a supervised classifier trained on leading indicators may identify transitions before they appear in realized volatility. *Difference: they use unsupervised generative HMM-GARCH; this project uses discriminative supervised classifiers with interpretable on-chain features.*

Ardia等人（2019）用马尔可夫转换GARCH模型分析比特币日收益率，发现比特币存在至少两个统计上显著不同的波动率regime。他们的发现支持了本项目的出发点：如果比特币市场本就有结构性阶段，那么用领先指标

（链上数据）提前识别这些阶段就有理论依据。区别：他们用的是无监督生成模型（HMM-GARCH），本项目用有监督判别分类器，输入特征可解释。

② On-chain + ML for price classification (Rafi et al., 2025)

Trains SVM, RF, Gradient Boosting, and deep learning models using 92 on-chain metrics + 138 technical indicators for Bitcoin price direction/magnitude prediction. Finding: on-chain data *significantly improves* classification over price-only baselines. Boruta-SVM produces the best risk-adjusted backtest returns. *Difference: this project uses a 3-class regime taxonomy, walk-forward evaluation, and an explicit on-chain ablation comparison.*

Rafi等人（2025）用92个链上指标+138个技术指标训练多种ML/DL模型预测比特币价格方向。核心发现：加入链上数据之后分类性能显著提升。这是本项目最直接的前置研究。区别：本项目用三分类市场阶段识别（而非涨跌二分类）、walk-forward评估、以及明确的价格-vs-链上消融实验。

③ ML classifier benchmark (Ballings et al., 2015)

Benchmarks ensemble methods (RF, AdaBoost, Kernel Factory) vs. single classifiers (NN, LR, SVM, KNN) for stock price direction across 5,767 European companies. Random Forest is the top algorithm. *This project replicates their model-hierarchy methodology but applies it to time-series multi-class classification with walk-forward evaluation.*

Ballings等人（2015）在5767家欧洲上市公司数据上系统比较了集成方法（随机森林等）与单一分类器（SVM、神经网络等）的股价方向预测效果，随机森林胜出。本项目借鉴他们的多模型层级比较框架，但应用于时序多分类任务+walk-forward验证。

6. Measure of Success

Primary: Macro-averaged F1 (regime classification)

- Macro-F1 treats all three classes equally regardless of frequency — directly penalizes ignoring the rare Distribution class.
- True baselines: stratified-random classifier yields macro-F1 ≈ 0.31 ; majority-class classifier ≈ 0.28 . All models are compared against these — not an arbitrary round number.
- Success targets: Linear SVM F1 ≥ 0.45 ; best nonlinear/neural model F1 ≥ 0.58 . Conservative, given genuine difficulty.

【主要评估指标】宏平均F1分数（三类平均，不按样本量加权）

- 为什么用宏平均F1：它平等对待三个类别，不允许模型靠只预测【扩张期】就得高分
- 真实基线：随机按类别比例预测的分类器宏F1 ≈ 0.31 ；只预测最多数类的分类器 ≈ 0.28 。所有模型与这两个基线比，不是对着凭感觉定的数字比
- 成功标准：线性SVM宏F1 ≥ 0.45 ；最好的非线性/神经网络模型 ≥ 0.58

Secondary: On-chain marginal value / ■■■■■■■■

- F1 delta: model (B) price+on-chain minus model (A) price-only, reported per fold. Consistent positive delta = evidence of on-chain signal. Null result = equally valid finding.

- 每个walk-forward fold单独报告 B-A 的F1差值。三个fold都正 = 链上数据有效；结论不稳定或为负 = 诚实报告为零结果

Regression sub-task / ■■■■■■30■■■■■■■

